

SESSION 2

From Stakeholders to Systems

Sustainability Systems, the Doughnut, and the Data Behind the Energy Transition

DEBRIEF: WHERE DID YOUR TEAM LAND?

After the presentations — three synthesis questions for the full team:

01

Where do you agree?

Find 2 claims all six analysts would accept — about Ørsted, about TotalEnergies, or about wind energy growth.

02

Where do you contradict?

Name one tension: where does the data analyst's numbers conflict with the foresight analyst's scenarios — or the systems analyst's feedback loops?

03

What is still missing?

What question did NO analyst answer well? This gap becomes your team's research priority for the next session.

Good consulting synthesizes partial views — it doesn't just list them.

THE CONTEXT: RAWORTH
(2017)

We chose the wrong goal.

"We have economies that need to grow whether or not they make us thrive. What we need are economies that make us thrive whether or not they grow."

— Kate Raworth, TED Talk, 2018

70+ years

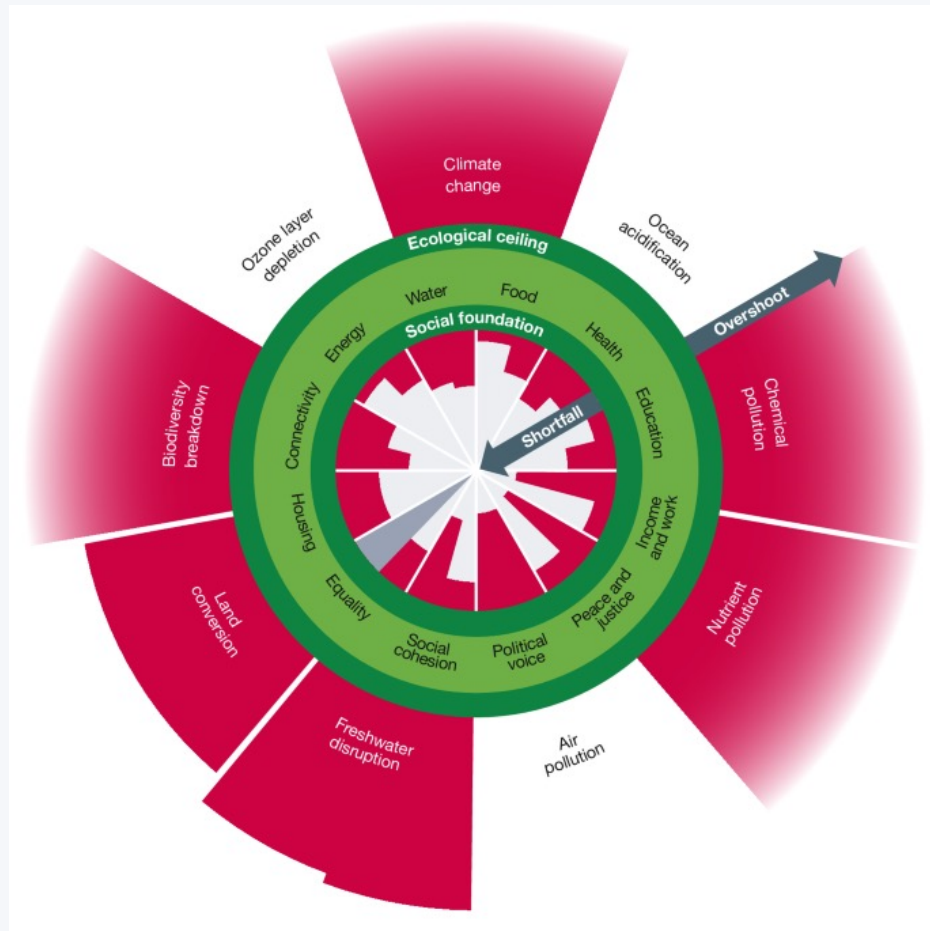
GDP as the dominant policy goal — since the 1940s

4 of 9

Planetary boundaries already breached while millions lack life's basics

The cuckoo metaphor: economics lost sight of its goals — GDP filled the empty nest.

THE DOUGHNUT: A 21ST-CENTURY COMPASS



Social Foundation (inner boundary)

12 life essentials every person deserves: food, water, energy, health, education, housing, income, political voice, gender equality, social equity, networks, peace & justice.

Ecological Ceiling (outer boundary)

9 planetary boundaries: climate change, biodiversity loss, land conversion, freshwater, ozone layer, ocean acidification, chemical pollution, air pollution, nitrogen & phosphorus loading.

The Goal: Dynamic Balance

Progress is not an ever-rising GDP line — it is humanity thriving between shortfall and overshoot, through regenerative and distributive economic design.

APPLYING THE DOUGHNUT: EUROPEAN WIND ENERGY

Where does wind energy sit on the Doughnut — and where does your consulting question live?

Ecological Ceiling Pressure

- Turbine manufacturing: rare earth minerals & steel
- Offshore cable infrastructure & seabed impacts
- Land use trade-offs for onshore wind
- End-of-life blade disposal (not yet solved)

Social Foundation Enablement

- Energy access & affordability for EU households
- Just transition: jobs in fossil vs. renewable sectors
- Energy security for smaller EU member states
- Local community acceptance of wind projects

Where Ørsted Sits

- Business model largely within the safe space
- Supply chain pressures push against the ceiling
- Offshore projects affect marine biodiversity
- Question: is it truly regenerative by design?

Where TotalEnergies Sits

- Core fossil business overshoots the ecological ceiling
- Renewables arm attempts to rebalance — but pace?
- LNG portfolio vs. wind growth targets: direct tension
- Can a company straddle both sides of the Doughnut?

THE KEY MOVE: FROM STAKEHOLDER MAP TO SYSTEM

1

Inventory — Who are the actors?

Ørsted, TotalEnergies, EU Commission, National TSOs, investment banks, household consumers, NGOs, regulators

Most Week 1 slides stopped here

2

Interactions — How do they influence each other?

Regulation → investment decisions → price signals → policy revision → new regulation → investment decisions...

Systems analysts need to reach this level

3

Dynamics — What are the feedback loops?

Reinforcing: falling LCOE → more investment → more scale → lower LCOE. Balancing: permitting bottleneck limits build-out despite falling costs.

This is where strategy lives

Raworth: the ecological ceiling is itself a feedback constraint — it is a balancing loop that caps economic activity whether we design for it or not.

WORKSHOP: DATA SCAVENGER HUNT — YOUR ANALYST ROLE

Each analyst: 20 minutes. Find your data. Do two sources agree?

DATA ANALYST

What share of EU electricity came from wind in 2024? Find it in Eurostat AND one other source.

Sources: Eurostat · IRENA · IEA Electricity Market Report

FORESIGHT ANALYST

What is the IEA projection for European offshore wind capacity in 2040 under different scenarios?

Sources: IEA World Energy Outlook · WindEurope · ENTSO-E TYNDP

ØRSTED ANALYST

What are Ørsted's stated LCOE targets and actual ROCE for offshore wind 2022 vs 2023?

Sources: Ørsted Annual Report · Bloomberg NEF · S&P Global

TOTALENERGIES ANALYST

What % of TotalEnergies capex went to renewables in 2023? What is the 2030 target?

Sources: TotalEnergies Sustainability Report · CDP Disclosure

SYSTEMS ANALYST

What is the grid curtailment rate for wind in Germany and Denmark? Where does this data live?

Sources: Bundesnetzagentur · Energinet.dk · ACER Annual Report

STRATEGY INTEGRATOR

Which of Raworth's five Doughnut factors (population, distribution, aspiration, tech, governance) is most visible in the EU wind case?

Sources: Synthesize from all above + Raworth Chapter 1

SOURCE QUALITY: KNOW WHAT YOUR DATA ACTUALLY IS

Not all data is equal — and a consultant who can't distinguish source types will be found out.

PRIMARY DATA

Examples: Company filings (annual reports, CDP, SEC 20-F), government statistical databases (Eurostat, Bundesnetzagentur), peer-reviewed datasets

✓ Highest — traceable to original source, audited or peer-reviewed

⚠ Publication lag of 1-2 years; self-reported elements exist

AGGREGATED STATISTICS

Examples: IRENA, IEA, WindEurope, EMBER — professionally compile and clean primary data for policy use

✓ High — widely cited by regulators and policymakers worldwide

⚠ Methodologies vary; definitions of 'renewable' differ across agencies

MODELED PROJECTIONS

Examples: IEA Net Zero scenarios, ENTSO-E TYNDP, BloombergNEF energy forecasts to 2040+

✓ Essential for foresight — scenario-dependent by design

⚠ Embed assumptions about policy, technology, behavior — always interrogate them

Before citing a number: Is this measured, compiled, or modeled? Which year? Which geography? Under what definition?

Your analyst role has a data question.

Before Session 3 (AI Tools for Sustainability Analysis), each analyst prepares a 1-page data memo:

1

Identify 3 datasets relevant to your analyst role's question from today's scavenger hunt.

2

Access or download at least one of them. Show the key figure or table.

3

Write 3 sentences: What does this data tell us? What does it NOT tell us? What question remains open?

Why: Session 3 is about AI tools for analysis — you need real data to bring into those tools. Come with data, leave with insight.

SESSION 2 COMPLETE



**THRIVE
IN
BALANCE**

"Ours is the era of the planetary household — and the art of household management is needed more than ever for our common home."

— Kate Raworth, Doughnut Economics (2017)